

A Frequentist Treatment of the Same Project

Household electricity demand, estimated by maximum likelihood

Project 17

October 10, 2026

Abstract

The assigned analysis is Bayesian. This note keeps the same household, the same daily series, the same regression, and the same six questions, and states how each answer is obtained from a maximum-likelihood fit instead of a posterior. The headline specification is unchanged: two annual harmonics, a linear trend, day-of-week effects, and an AR(1) error with Student- t innovations, on the kWh scale. With 1,417 days and a weak prior, the point estimates sit next to the posterior means already reported. What changes is the language of the claim: a confidence interval and a p -value, rather than a posterior probability.

1 What stays the same

The data preparation does not move. One-minute power from the UCI *Individual Household Electric Power Consumption* series is converted to daily energy by summing `Global_active_power` and dividing by 60. A day is kept only when at least 95% of its 1,440 minutes are observed. That leaves 1,417 usable days on a gap-free calendar of $T = 1,440$ days, from 17 December 2006 through 25 November 2010, with 23 days retained as missing so that the autoregression is defined on every date.

The mean structure is the one the brief requires. For a response $z_t = g(y_t)$, with g either the identity or the logarithm,

$$z_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \tag{1}$$

$$\mathbf{x}_t^\top \boldsymbol{\beta} = \beta_0 + \beta_1 \frac{t - \bar{t}}{365} + \sum_{j=1}^J \left[a_j \sin \frac{2\pi j t}{365} + b_j \cos \frac{2\pi j t}{365} \right] + \sum_{k=2}^7 \delta_k \mathbf{1}\{\text{dow}_t = k\}, \tag{2}$$

$$\varepsilon_t = \phi \varepsilon_{t-1} + \eta_t, \quad \eta_t \sim t_\nu(0, \sigma^2). \tag{3}$$

Monday is the baseline day. The trend is centred at $\bar{t}$, so β_0 is the level at the midpoint of the record. The harmonics use raw t , as mandated. The specification that the holdout selected, called M7 in the main report, is $J = 2$, AR(1), t innovations, and g the identity, so z_t is daily consumption in kWh.

The twelve-month holdout is also unchanged. Models are fitted on data through 30 November 2009 (1,072 observed days) and scored on the final 345 observed days. Log score, CRPS, and empirical coverage do not depend on a posterior. They remain the right way to decide whether a specification forecasts.

2 What the fit is

There is no prior. The parameters $(\boldsymbol{\beta}, \phi, \sigma, \nu)$ are fixed unknown constants, and the fit is the value that maximises the likelihood of the observed days.

For Gaussian innovations the conditional likelihood, after the AR(1) quasi-difference $\tilde{z}_t = z_t - \phi z_{t-1}$ and $\tilde{\mathbf{x}}_t = \mathbf{x}_t - \phi \mathbf{x}_{t-1}$, is ordinary weighted least squares in β . In R the Gaussian special case is

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arima(y, order = c(1, 0, 0), xreg = X, method = "ML").
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The exact likelihood adds the stationary density of the first residual, with variance $\sigma^2/(1 - \phi^2)$. That is the same initial-error term the Gibbs sampler uses; here it enters the likelihood once, at the maximum, rather than at every draw.

Student- t innovations use the same normal scale mixture as the sampler,

$$\eta_t \mid \lambda_t \sim N(0, \sigma^2/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2),$$

but as an EM algorithm.

- **E-step.** Conditional on a current innovation value, $\hat{\lambda}_t = (\nu + 1)/(\nu + \eta_t^2/\sigma^2)$. A small weight is a day the likelihood downweights. These weights are the conditional expectations of the latent scales, evaluated once at the maximum likelihood estimate. For a missing day, the E-step also averages over its conditional state distribution.
- **M-step.** Weighted least squares for β , then updates for σ^2 , ϕ , and ν .

The algorithm stops at one point $(\hat{\beta}, \hat{\phi}, \hat{\sigma}, \hat{\nu})$ and one set of weights $\hat{\lambda}_t$.

The 23 missing days are integrated out of the observed-data likelihood. A Kalman filter does this exactly in the Gaussian case; for the Student- t scale mixture, EM or Monte Carlo EM averages over the missing states and latent scales. Simply filling a missing day with its conditional mean and treating it as observed would understate uncertainty. A parametric bootstrap then carries missing-state uncertainty into the standard errors. The Bayesian fit samples the missing states inside the sampler, so that uncertainty is already inside every posterior draw.

Standard errors come from the observed Fisher information at the maximum likelihood estimate. A function of the coefficients — an amplitude, a phase, the weekend contrast, a percent-per-year slope, the horizon factor $1/\sqrt{1 - \phi^2}$ — is handled by the delta method. When that approximation is poor, as it is for an amplitude near zero or a phase near a branch cut, the interval is read off a parametric bootstrap instead.

3 The six questions

The numerical anchors below are the posterior summaries already computed for M7. Under a weak prior they are a close guide to the maximum likelihood estimates. Each paragraph says what the frequentist report would quote.

3.1 Q1 — variation over the study period

The midpoint level, the seasonal range, and the one-step innovation standard deviation are maximum likelihood estimates, with confidence intervals from the information matrix or the bootstrap. In the Bayesian fit those figures are a midpoint level of 25.96 kWh/day, a full seasonal range of 18.25 kWh/day, a weekly contribution of about 4.56 kWh, and an innovation standard deviation of 5.59 kWh. The ordering is the same one the exploratory R^2 decomposition already gave: the first harmonic carries almost all of the systematic variance (R^2 increment 0.36), day of week adds 0.05, and the trend adds 0.01.

The Newey–West intervals in the exploratory note are a descriptive frequentist summary that ignores the AR(t) structure. The model-based analysis replaces them with intervals from the fitted autoregression.

3.2 Q2 — long-term trend

The slope estimate is -0.186 kWh/year. The 95% interval already computed, $[-0.667, +0.292]$, implies a standard error of about 0.245, a z -statistic of about -0.76 , and a two-sided p -value for $H_0 : \beta_1 = 0$ of about 0.45. The interval covers zero. Four years do not establish a trend, and the implied change over the whole record is under 3% of the mean level.

The Bayesian report also quotes $\Pr(\beta_1 < 0 \mid \text{data}) = 0.775$. That number is the posterior mass below zero. With a flat prior and a normal sampling distribution it is $\Phi(0.76) \approx 0.78$, which is why the two analyses look so similar. The frequentist summary stops at the estimate, the confidence interval, and the test: the point estimate is negative, and a test of a zero slope does not reject at conventional levels.

3.3 Q3 — annual pattern, and the second harmonic

Amplitude and peak date are

$$\hat{A}_j = \sqrt{\hat{a}_j^2 + \hat{b}_j^2}, \quad \widehat{\text{phase}}_j = \text{atan2}(\hat{a}_j, \hat{b}_j),$$

with delta-method or bootstrap intervals. In the Bayesian fit the first harmonic has amplitude 8.08 kWh and peaks on 14 January; the second has amplitude 2.43 kWh, and the amplitude ratio is 0.30.

Harmonic 2 against harmonic 1 is a likelihood-ratio test on two extra coefficients, referred to χ^2_2 , together with AIC. The comparison is nested and regular. The substantive reason to keep the second harmonic is unchanged: it deepens and delays the summer trough and sharpens the winter peak. The Bayesian statement $\Pr(A_2 > 0.15 A_1 \mid \text{data}) = 0.999$ becomes a confidence interval for the ratio that lies well above 0.15.

3.4 Q4 — weekly pattern

Each day-of-week deviation has a Wald interval from $\widehat{\text{Var}}(\hat{\beta})$. The weekend-minus-weekday contrast is the linear combination $\mathbf{c}^\top \hat{\beta}$, with standard error $\sqrt{\mathbf{c}^\top \widehat{\text{Var}}(\hat{\beta}) \mathbf{c}}$. The Bayesian fit puts Saturday and Sunday about 3.5 kWh and 3.0 kWh above the weekly mean, and the weekend-minus-weekday contrast at 4.56 kWh/day with interval $[3.71, 5.42]$. The posterior probability that the weekend exceeds the weekday is 1. The frequentist report is the same contrast, a confidence interval lying entirely above zero, and a very small p -value for a zero contrast.

3.5 Q5 — accuracy, and how uncertainty grows with the horizon

The h -step point forecast plugs the maximum likelihood estimate into the same recursion the sampler uses,

$$\hat{z}_{T+h} = \mathbf{x}_{T+h}^\top \hat{\beta} + \hat{\phi} \hat{\varepsilon}_{T+h-1},$$

with future errors propagated from the last observed residual. For Gaussian innovations a plug-in prediction interval uses the scale

$$\hat{\sigma} \sqrt{\sum_{j=0}^{h-1} \hat{\phi}^{2j}}.$$

For t innovations, a one-step interval can use $t_{\hat{\nu}}$ quantiles. At $h > 1$, however, the weighted sum of independent Student- t innovations is not itself Student- t , so the predictive quantiles must be obtained by simulating complete future paths. To include parameter-estimation uncertainty, a parametric bootstrap simulates and refits a training series before producing each set of forecast paths.

With $\hat{\phi} = 0.375$ the width rises by about 8% and then stops. The ceiling is the stationary factor $1/\sqrt{1 - \hat{\phi}^2} \approx 1.08$. In the Bayesian fit the 95% predictive interval widens from 28.9 kWh at one day ahead to about 31 kWh by day seven, and the empirical coverage is 94.5% at $h = 1$ and 95.2% at $h = 7$. The frequentist calculation reproduces that shape. A plug-in interval treats $(\hat{\beta}, \hat{\phi}, \hat{\sigma}, \hat{\nu})$ as known, so it is slightly narrow. A parametric bootstrap of whole forecast paths puts the estimation error back in. Knowing yesterday's consumption still buys only a few days of sharper intervals. Beyond about a week the forecast is the seasonal-plus-weekly mean.

3.6 Q6 — periods the mean does not explain

One-step anomalies are tails of the fitted innovation distribution at the maximum likelihood estimate: a day is flagged when its conditional residual is extreme under $t_{\hat{\nu}}(0, \hat{\sigma}^2)$. A one-step residual cannot see a level shift that the AR(1) has already absorbed, so a sustained anomaly is still judged by a block statistic. The same 14-day mean used in the main report is referred to the distribution of that mean under the fitted stationary process, by simulation from the maximum likelihood estimate or by a parametric bootstrap.

The episodes do not change. The August blocks are the house running near a base load through the French summer holiday, and the winter blocks are high days the seasonal mean does not reach. Both are behavioural or meteorological, and neither is a function of the calendar terms in (2).

4 Choosing among the seven specifications

Nested Gaussian comparisons use likelihood-ratio tests and AIC. AR(2) against AR(1) adds one coefficient. $J = 2$ against $J = 1$ adds two. The logarithm against the identity is not a nested comparison; both maximised likelihoods have to be densities for daily kWh, so the log-scale fit contributes an extra $-\log y_t$. Gaussian innovations against t innovations are compared by AIC, because $\nu \rightarrow \infty$ sits on the boundary of the parameter space.

The holdout designs carry over as written. In the Bayesian comparison, WAIC preferred AR(2) and the twelve-month forecast rejected it, because the extra lag was fitting the absence blocks one step ahead. AIC can make the same mistake. The forecasting question still selects M7: two harmonics, AR(1), t innovations, kWh scale. The log transform remains the worse forecasting scale.

5 Bayesian and frequentist approaches side by side

The two approaches can use exactly the same response, design matrix, AR(1) structure, Student- t innovations, missing-day rule, and holdout period. The difference is how uncertainty about the unknown parameters is represented and interpreted. The table makes that distinction for this model rather than in the abstract.

Part of this project	Bayesian analysis in the main report	Frequentist version in this note
Unknown parameters	After the data are observed, uncertainty about $\theta = (\beta, \phi, \sigma, \nu)$ is represented by the posterior $p(\theta y)$. A statement such as $\Pr(\beta_1 < 0 y) = 0.775$ is allowed, conditional on the model and priors.	θ is a fixed but unknown value. Randomness refers to data that could have been generated repeatedly from the model. The analysis does not assign a probability that the realised fixed β_1 is negative.

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Part of this project	Bayesian analysis in the main report	Frequentist version in this note
Information used	The likelihood from the $AR(1)+t$ model is combined with the stated weakly informative priors for β , σ , ϕ , and ν .	The observed-data likelihood and parameter constraints are used, but no prior distribution is supplied. Stationarity ($ \phi < 1$) and $\nu > 2$ are model constraints, not priors.
What is computed	Gibbs and Metropolis sampling produce draws from the joint posterior, including draws of missing daily demand and the latent t -mixture weights. A posterior mean is a point summary of those draws.	Likelihood or EM optimisation produces one maximum likelihood estimate $\hat{\theta}$. Curvature of the likelihood or a parametric bootstrap gives standard errors; missing values and robust weights may appear as intermediate EM quantities.
95% parameter interval	A 95% credible interval contains 95% of the posterior probability for the parameter, conditional on this dataset, this likelihood, and these priors.	A 95% confidence procedure covers the fixed true parameter in 95% of repeated datasets generated under the model. After one interval is computed, it is not interpreted as a 95% probability that the fixed parameter lies inside it.
Q2: long-term trend	Report the posterior of β_1 : mean -0.186 kWh/year, 95% credible interval $[-0.667, 0.292]$, and $\Pr(\beta_1 < 0 y) = 0.775$. The posterior says a decline is more plausible than an increase, but the uncertainty is wide.	Report $\hat{\beta}_1$, its 95% confidence interval, and a test of $H_0 : \beta_1 = 0$. Using the reported interval as a close approximation gives $z \approx -0.76$ and two-sided $p \approx 0.45$, so zero trend is not rejected. The p -value is not the probability that H_0 is true.
Q3: second annual harmonic	Transform every posterior draw of (a_j, b_j) into amplitude and phase. Report the posterior interval for A_2/A_1 and $\Pr(A_2 > 0.15A_1 y) = 0.999$, then check WAIC and holdout forecasting.	Estimate amplitude and phase from $(\hat{a}_j, \hat{b}_j)$. Test the regular null $H_0 : a_2 = b_2 = 0$ with a two-degree-of-freedom likelihood-ratio test; use a bootstrap interval for A_2/A_1 , AIC, and the same holdout forecasts.
Q4: weekly pattern	Compute each day effect and the weekend-minus-weekday contrast for every posterior draw. The contrast is 4.56 kWh/day with posterior interval $[3.71, 5.42]$ and posterior probability of a positive contrast essentially one.	The same contrast is a linear combination $\mathbf{c}^\top \hat{\beta}$. Its standard error is $\sqrt{\mathbf{c}^\top \widehat{\text{Var}}(\hat{\beta}) \mathbf{c}}$. Report a Wald or bootstrap confidence interval and a test of a zero contrast.
Heavy tails and unusual days	Each latent weight λ_t has a posterior distribution. Small posterior mean weights identify days downweighted by the Student- t likelihood, while posterior uncertainty in ν describes how heavy the tails are.	Evaluate the robust weight $\hat{\lambda}_t = (\hat{\nu} + 1)/(\hat{\nu} + \hat{\eta}_t^2/\hat{\sigma}^2)$ at the MLE. Small values flag downweighted days. Uncertainty in ν or a weight is obtained from likelihood profiling or a bootstrap.
The 23 missing days	Missing daily values are latent variables sampled with the parameters. Their posterior uncertainty automatically flows into coefficient summaries and forecasts.	Integrate the missing states out of the observed-data likelihood, or use the conditional-Gaussian scale-mixture representation inside EM. Smoothed values are estimates, not observed data; a parametric bootstrap propagates their uncertainty.

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Part of this project	Bayesian analysis in the main report	Frequentist version in this note
Q5: future demand	The posterior predictive distribution averages forecasts over posterior draws of θ and over future Student- t innovations. Thus parameter and future process uncertainty are both included directly.	A plug-in forecast substitutes $\hat{\theta}$ and includes future innovation variance. A parametric bootstrap is needed to add parameter-estimation uncertainty. A 95% prediction procedure is judged by repeated-sample coverage; future demand itself is random in both approaches.
Forecast-horizon result	Posterior draws give $\phi = 0.375$ with interval $[0.323, 0.426]$ and imply an uncertainty ceiling near $1/\sqrt{1 - \phi^2} = 1.08$. The predictive interval therefore stops widening after roughly four days.	Insert $\hat{\phi}$ into the variance multiplier $\sqrt{\sum_{j=0}^{h-1} \hat{\phi}^{2j}}$. This is the ratio to one-step innovation uncertainty for any independent, finite-variance innovations. Simulate the Student- t predictive quantiles, and use a delta-method or bootstrap interval for the ceiling. The same estimated ϕ produces the same saturation after roughly four days.
Q6: anomalous periods	Posterior predictive tail areas compare each observed day, and each rolling 14-day mean, with replicated values averaged over posterior uncertainty.	Tail areas are calculated at $\hat{\theta}$, or calibrated by parametric bootstrap. The same 14-day block statistic is essential because one-step residuals miss a sustained shift once AR(1) has absorbed it.
Choosing a model	WAIC compares expected one-step predictive fit; the held-out log score, CRPS, and coverage decide whether the choice generalises. The holdout overrides WAIC when they disagree.	Likelihood-ratio tests compare regular nested models, and AIC compares maximised predictive fit. The same held-out log score, CRPS, and coverage remain decisive. Log-scale likelihoods must include the $-\log y_t$ Jacobian to be comparable on the kWh scale.
What should be similar	With 1,417 days and weak priors, posterior means and credible intervals for well-identified regression effects should be close to likelihood estimates and confidence intervals.	The MLE should therefore support the same broad conclusions. Larger differences are most likely for weakly identified or nonlinear quantities such as ν , phase, missing-day values, and extreme-tail probabilities.
Shared limitation	Both answers are conditional on the AR(1)+ t model and the available calendar predictors. Neither approach repairs omitted weather or occupancy information. The twelve-month holdout is therefore essential for both.	

The shortest distinction is this: Bayesian probability describes uncertainty about the parameters *after conditioning on these data and the priors*; frequentist confidence describes the long-run calibration of an estimation procedure over hypothetical repeated datasets. Neither interpretation makes one fitted model automatically more accurate, which is why the same external holdout remains the final check.

6 What the conclusions still say

The annual cycle dominates. The second harmonic is needed because the summer trough is deeper and later than a single sine wave. The weekend premium is several kWh per day, and its confidence interval lies entirely above zero. A linear trend is not resolved in four years. Forecast uncertainty stops widening after about four days, because ϕ is only about 0.4. Beyond a week, a forecast for

this house is seasonal-plus-weekly climatology.

The assignment asks for posterior probabilities, and those statements stay in the main report. This note records the sampling-theory version of the same fit, for the same questions, on the same series.